What Was Searched

Five months of systematic strategy research on CME futures — the complete record of every family tested, how large each search was, and what it returned.

27.55B	48	23	1
configurations evaluated	families closed	markets	survivor

Every number in this document is read from the repository's own result files and its append-only hypothesis ledger. Nothing is estimated. The single most important conclusion is not any individual result but the arithmetic in section 5: **a search of 27.55 billion configurations demands 7.73 sigma of evidence, and nothing in finance clears that bar.** The largest searches in this project could not have found a real edge even while looking directly at one.

1 The scale

Campaign	Configurations	Cleared gate	Survived
Megatick — tick-event configurations	26,131,683,248	—	0
FX / AND-combos of technical features	1,379,701,482	—	0
Feature search (FSEARCH + F2, 6 markets)	33,142,022	13,785,856	146, then 0
Needle — 1.2M strategies, all 8 contracts	1,209,600	—	0
Structural strategies (103,680 space)	81,348	—	0
Targeted searches (final phase)	13,433	—	1
TOTAL	**27,545,831,133**		**1**

The feature-search family alone scanned 33.1 million candidates; 13.8 million cleared an in-sample gate, 1.28 million also paid on a held-out 40%, and **146 survived every quarter** — of which none survived a shuffled control. That funnel is the shape of the whole project.

2 Phase one — the brute-force era (35 ledger families)

An append-only ledger recorded each family so no later search would rediscover a corpse. Verdicts are quoted from it.

#	Family	Scale	Verdict
1	Impulse-pullback, all variants	1000s of configs	NULL
2-7	Mean reversion, VWAP, opening range, squeeze, gap, volume spike	grids	NULL
8	Hour-of-day / weekday / weekday x hour	grid	NULL
9	Cross-market lead-lag	grid	NULL
10	Order-flow imbalance (trade prints)	grid	NULL
11-13	Position sizing, calendar effects, sweeps/absorption	grids	NULL
14	Trailing-stop exits (13 rules)	full	NULL
15	AND-combos of technical features	1.38B configs	NULL
16	Daily GEX / options positioning	daily	NULL
17	Book imbalance	IC study	REAL, small
18	COT / positioning	weekly	NULL
19	Selection-by-train-score as a method	1.38B	HARMFUL
20	FX impulse at 97% lower costs	full sim	NULL
21	Leg-grammar conditional cells	~750 cells/scale	RETRACTED — tape artifact
22	Back-of-queue market making	full day	NEGATIVE — maker path closed
23	**Megatick tick-event configurations**	**26,131,683,248**	NULL
24	Selectivity: are runaway swings visible in advance?	13 features x 5 quintiles	REAL, far too small
25	Exit targets instead of waiting for the turn	45 combinations	NULL
26	Brackets: first-touch on the real tick path	56 shapes x 2 directions	NULL — decisive

#	Family	Scale	Verdict
27	Cost reduction by switching instrument (bonds)	—	WRONG — withdrawn
28	CME order-book data as the remaining hope	—	CLOSED without buying
29	Structural targets: prior swing highs/lows as attractors	268k / 125k / 46k trades	REAL but INVERTED
30	103,680 structural strategies	81,348 scored	NULL
31	Symmetric-barrier directional asymmetry	+/-24, 48, 96 ticks	NULL — retracted same session
32	conf_len — top result of the 103,680 search	8 strategies x 2 splits	NULL — pure selection noise
33	LOCATION: ten pre-named destinations	14 rows x 2 splits	NULL
34	Needle: 1.2M strategies, pass in ALL 8 contracts	1,209,600 x 2 arms	NULL — 123x worse than control
35	**THE CEILING**: gradient-boosted model over the whole feature set	4 horizons x 2 arms, purged	**ZERO** — bounds every other result

3 Phase two — 13 targeted searches

After the brute-force era closed, the method changed: far fewer configurations, each aimed at a region the arithmetic said could work, every one reporting the smallest effect it could have detected. Thirteen searches, 13,433 configurations in total.

Search	What it tested	Configs	Result
cadence	Trade the 18-hour signal once instead of 400 times	24	–$573 → **+$246/wk**
perm400	20 independent permutations at h=400	21	**p = 0.048**, IC 3.0 se
tradestats	Trade-level statistics on one micro	3	Edge sat in the WEAK signals
riskcap	Does it survive a hard dollar stop?	8	Survived (–18%), DD unchanged
ablate	Which feature family carries the result?	6	**KILLED IT**
xsec	Cross-sectional ranking, 30 min to 3 days	168	0 winners, max |t| 1.64
wide	18 markets x 3 horizons, cross-market ML	108	0 of 54; NG 4.4 sigma but $58/wk
trend	Trend following, 4 to 12 week holds	48	Calmar 1.08 — MDE $645 vs $271
lowdd	Max $2,000 drawdown at 20 micros	72	0 of 72; floor was $15,504
mindd	The full drawdown frontier	144	Every point below its noise floor
daily_stats	Daily marks at 2 minis	2	Sign FLIPPED on longer data
lowfreq	Calendar, session, COT, volatility regime	318	**1 survivor**
brackets_wide	45% win at 1:2, daily-scale brackets	12,672	2 met the brief, 0 survived
fib_verify	Independent rebuild of a handed-over strategy	6	Shuffle BEAT it

4 The one survivor — overnight drift

Buy the regular-session close, sell the next regular-session open. One trade a day, no intraday exposure.

Market	$/trade	t	Win rate	Same market, INTRADAY
Gold (GC)	+$38.2	2.58	58.9%	–1.12
Nasdaq (NQ)	+$29.2	2.01	56.5%	+0.04
Copper (HG)	+$10.3	1.60	53.0%	–0.73
S&P (ES)	+$10.6	1.49	56.1%	+0.60
Russell (RTY)	+$3.4	0.87	54.1%	–0.11

Pooled book	$/week	t	Win rate	Max DD	Calmar
Equity + metals, OVERNIGHT	**$304**	**2.20**	57.2%	$10,282	1.54
the same book, intraday	–$54	–0.37	53.1%	$16,136	–0.18

Why this one and nothing else. It is the only result in five months with all three properties: a published mechanism rather than a found pattern; agreement across correlated markets (the test that killed the NQ breakout, which scored 3.86 sigma on Nasdaq and 0.17 on the S&P); and a control that fails in the *right* direction — the same markets are flat to negative during the day session. It is also stable year by year: $310, $311 and $304 per week in 2024, 2025 and 2026, so it is not a recent artifact.

What it is not. t = 2.20 against a 4.19 sigma bar for the 318 configurations it came from. Four years is not enough to settle it. And because it holds positions overnight, it cannot be run inside a funded account, which requires flat by the close.

5 The finding that reframes all of it

The significance bar rises with the number of things tried, as the square root of twice the log of the count. Every extra configuration raises the evidence bar for *all* of them.

Search size	Bar to clear	Would it accept a real 4.4 sigma edge?
54 configurations	3.62 sigma	YES
12,672 configurations	5.15 sigma	no
1,209,600 configurations	6.32 sigma	no
26,131,683,248 configurations	**7.73 sigma**	**NO**

Natural gas measured **4.36 sigma** in this project — the strongest statistical result found. Discovered inside a 54-config search it clears comfortably. The identical result inside the 26-billion sweep **fails**. Same signal, same data, same strength; accepted or rejected purely by how much else was looked at alongside it.

Brute force did not merely waste time. It destroyed the ability to detect an edge. Twenty-six billion configurations could not have found natural gas even while looking straight at it. Thirteen thousand aimed configurations found more real signal than twenty-seven billion unaimed ones.

6 The second blind spot — measured power

Edges of known size were planted in the engine's own data and it was asked to find them:

Planted edge	Worth, per week	Detected
+0.05 round trips / trade	~$80	**0%**
+0.50 round trips / trade	~$800	**0%**
+2.00 round trips / trade	~$3,200	60%

An edge worth **$800 a week was invisible 100% of the time**. So “nothing found” never meant “nothing is there” — it meant “nothing large enough for this search to see”. The repository's own calibration module had said so all along: *“its silence was never evidence of absence; it was evidence of no power.”* Every search in the final phase now prints that number beside its result, found or not.

7 Why the short-horizon work could never pay

Holding period	Average NQ move	Share eaten by $1.99 cost	Must forecast
1 minute	$36	5.5%	5.5% of the move
5 minutes	$81	2.4%	2.4%
30 minutes	$195	1.0%	1.0%
2 hours	**$371**	**0.54%**	**0.54%**

Essentially all 27.5 billion configurations were tested between 15 seconds and 50 minutes — the region where the toll is largest. Cost is fixed per trade while the move grows with time, so the problem was never commission. It was trading quickly. Execution work (cheapest broker, limit orders, no slippage) takes $1.99 down to about $1.44 against a break-even of $0.70: still twice over, and a dead end in that direction.

8 What remains

Lane	Status	What would settle it
Short-horizon price patterns	Closed	27.5B configurations. Decisively dead.
Maker / passive fills	Closed	6.6% measured fill rate on MNQ.
Cross-sectional ranking	Closed	168 configs with power behind the null.
Cross-market ML at 18 hours	Closed	A shuffled target out-earned it.
Positioning (COT, 9 years)	Closed	Shifted control beat the real signal.
Trend following	Unresolved	Needs ~15 years; 4.2 on disk. Free download.
Overnight drift	**Alive**	**Testable to the 1990s with free daily data.**

The single next step

Three of the four remaining questions have the same answer: **long price history**. Trend following needs about fifteen years and has four. Overnight drift is testable back to the 1990s. The Nasdaq breakout could be settled with daily data to 2010. All three are free downloads rather than discoveries. After 27.55 billion configurations, the binding constraint is no longer ideas or compute — it is that two to four years of tape cannot tell a real effect from a lucky one.

Compiled from the repository's result files (CADENCE, PERM_H400, ABLATE_H400, XSEC, WIDE, TREND, LOWDD, MINDD, DAILY_STATS, LOWFREQ, BRACKETS_WIDE, FIB_VERIFY) and the append-only hypothesis ledger. Costs throughout are the measured $1.99 per MNQ round turn taken from real fills, never a hopeful figure.